

Phase 17 Demo: Semicolon-Separated Bindings

Example 1 : Two bindings

Simple case with two binding groups:

$$\forall x : \mathbb{N}; y : \mathbb{N} \bullet x + y > 0$$

Example 2 : Three bindings

Three levels of nesting:

$$\forall x : T; y : U; z : V \bullet x = y$$

Example 3 : Mixed comma land semicolon

First quantifier has multiple variables with shared domain:

$$\forall x, y : \mathbb{N}; z : \mathbb{N} \bullet x + y = z$$

Example 4 : Real - world example from Solution 41

$$\forall r : \text{ran } records \bullet \forall i1, i2 : \text{dom } r \bullet i1 \neq i2$$

Example 5 : Conditional with semicolon bindings

Combining Phase 16 and Phase 17:

$$\forall x : \mathbb{N}; y : \mathbb{N} \bullet \text{if } x > y \text{ then } x \text{ else } y$$

Example 6 : Nested mu operator

Definite description with multiple bindings:

$$(\mu p : \text{ran } hd; q : \text{ran } hd \mid p \neq q \bullet p.2 > q.2)$$

Example 7 : With exists quantifier

Multiple bindings with exists:

$$\exists x : \mathbb{N}; y : \mathbb{N} \bullet x > y \wedge y > 0$$